WACV 2027 UPAR Challenge
Fact sheet

This is the fact sheet's template for the WACV 2027 UPAR Challenge. Please fill out the following sections carefully in a scientific writing style. Then, send the compressed project (in `.zip` format), i.e., the generated PDF, `.tex`, `.bib` and any additional files to `upar.challenge@gmail.com`.

I. TEAM DETAILS

- Team leader name: Kaveh Safavigerdini
- Username(s) on Codabench: kavehsfv-957256-957256_VID-Docker-GPU
- Team leader affiliation: Electrical Engineering and Computer Science Department, University of Missouri - Columbia, Mo, USA
- Team leader email: ksgh2@umsystem.edu
- Name of other team members (and affiliation): —
- Team website URL (if any): —
- Competition track (mark with X one single option)[1]:
  - (X) Track 1: Pedestrian Attribute Recognition.
  - ( ) Track 2: Attribute-based Person Retrieval.
  - ( ) Track 3: Human Pose Estimation.

II. CONTRIBUTION DETAILS

A. Title of the contribution

Fine-tuning a CLIP ViT-B/16 visual encoder with class-balanced weighted BCE for unified 40-attribute pedestrian attribute recognition.

B. Representative image / workflow diagram of the method

The pipeline is deliberately minimal. A pedestrian crop is letterboxed to 224×224 and normalized with CLIP statistics; the frozen ImageNet-style preprocessing is followed by the fine-tuned CLIP ViT-B/16 visual tower (12 transformer blocks, width 768, QuickGELU MLPs, class-token pooling, learned 512-d projection); a linear head maps the pooled embedding to 40 attribute logits; a fixed per-attribute bias vector (calibrated on the validation set) is added before the sigmoid to produce the final probabilities. The Codabench submission applies these probabilities directly with the fixed 0.5 threshold.

C. Detailed method description

Architecture. We use the visual tower of OpenAI's CLIP ViT-B/16 (patch size 16, 12 blocks, embedding width 768, 12 heads, QuickGELU activations, class-token pooling followed by a learned projection to 512 dimensions), initialized from the public "openai" weights [1]. A single linear layer maps the 512-d pooled image embedding to 40 attribute logits. The model is fully self-contained in the submission (plain PyTorch re-implementation of the ViT, no external dependencies beyond PyTorch and Pillow).

Training objective. Training uses binary cross-entropy with a per-attribute positive weighting to counter the severe label imbalance across the 40 attributes (positive rates range from 0.2% to 96.5% on the training split). For attribute a with training positive rate p_a, the pos-weight is

$$w_a = \left(\frac{1 - p_a}{p_a}\right)^{0.5}, \tag{1}$$

i.e., the square root of the inverse label ratio, which tames the extreme weights of the rarest attributes while still up-weighting them by up to two orders of magnitude.

Optimization. AdamW with weight decay 0.05, differential learning rates (backbone 2×10^{-5}, classification head 1×10^{-3}), 500-iteration linear warmup followed by cosine decay, batch size 256 (accumulated to effective batches of 512 on a single 32 GB GPU), 6 epochs, automatic mixed precision (bf16), and gradient-norm clipping at 1.0.

Data. Training used only the predefined UPAR training split (97,669 crops: 10,000 Market1501, 79,001 PA100k, 8,668 PETA). The provided validation split was *not* used for gradient updates; it was used only for checkpoint selection, per-attribute bias calibration, and local metric tracking.

Post-processing. A per-attribute additive bias b_a in logit space is tuned by coordinate ascent on the validation set to maximize the challenge metric (harmonic mean of mA and Instance_F1). At inference the model returns $\sigma(z_a + b_a)$. This lifted validation Challenge_Avg from 0.8751 to 0.8835 (mA 0.8717→0.8941) with a small Instance_F1 trade-off (0.8784→0.8731).

Checkpoint selection. The submitted weights are the epoch-2 checkpoint, selected on validation Challenge_Avg (0.8751 raw, the best of all epochs of the final run).

D. Challenge results

Results of the latest submission on the Codabench leaderboard (test set, 36,343 images):

Metric	Score
Challenge_Avg	0.7599
mA	0.7716
Label_F1	0.5341
Instance_Acc	0.6257
Instance_Prec	0.7152
Instance_Rec	0.8009
Instance_F1	0.7486

[1] If you participated in more than one competition track, you need to share with the organizers one fact sheet per track.

E. Final remarks

Pros. (i) A single, compact fine-tune of a strong general-purpose visual encoder reaches competitive accuracy without any PAR-specific architecture. (ii) The weighted-BCE exponent (0.5) and the validation-calibrated per-attribute biases give robust per-attribute balance, which benefits mA and Label_F1 on rare attributes. (iii) The whole system is a self-contained plain-PyTorch file plus one weight file, which makes verification straightforward.

Cons. (i) The gap between validation (Challenge_Avg 0.8835) and test (0.7599) indicates distribution shift between the splits and/or mild over-calibration to the validation set; the bias calibration in particular may have overfit the validation distribution. (ii) No ensemble or test-time augmentation was used; both typically add 1–2 points. (iii) Rare attributes (e.g., purple/yellow/orange lower-body clothing colors) remain the dominant error source.

III. Additional method details

Please, reply if your challenge entry considered (or not) the following strategies and provide a brief explanation. For each question, mark with X one single option.

- **For the competition track associated with this fact sheet, you confirm that you have trained your model on the predefined datasets** (X) Yes, ( ) No
  Training used only the predefined UPAR training split (Market1501 + PA100k + PETA crops, 97,669 images). No other pedestrian data was used for training.
- **Did you use any pre-trained model:** (X) Yes, ( ) No.
  If yes, please detail (e.g., model architecture, training strategy, train data, etc.).
  The visual tower of OpenAI CLIP ViT-B/16 (public "openai" weights via open_clip_torch) was used as initialization. CLIP was trained on Web-scale image–text pairs (WIT-400M); we use only its ImageNet-like visual encoder weights, fine-tune all layers, and do not use any text encoder, prompts, or language supervision at any stage of our pipeline.
- **Did you use external data?** ( ) Yes, (X) No
  If yes, please detail:
  Only the CLIP initialization weights (a foundation-model prior, not PAR-specific data). No pedestrian-attribute datasets, pseudo-labels, or additional images beyond the provided splits were used.
- **Did you perform any data augmentation?**
  (X) Yes, ( ) No
  If yes, please detail:
  Random horizontal flip (p=0.5) and random erasing of a grey rectangle (p=0.25, rectangle size between 1/8 and 1/3 of each side), applied after letterbox resizing. The grey value equals the normalized zero (mean) value, matching padding semantics.
- **Did you use the provided validation set as part of your training set?** ( ) Yes, (X) No
  If yes, please detail:
  The validation set was never used for gradient updates. It was used for (i) checkpoint selection (best epoch by Challenge_Avg), (ii) per-attribute bias calibration (coordinate ascent on 40 bias values), and (iii) local metric reporting. All gradients came exclusively from the 97,669 training crops.
- **Did you use any regularization strategies/terms?** ( ) Yes, (X) No
  If yes, please detail:
  Weight decay 0.05 (AdamW) on all parameters; gradient-norm clipping at 1.0; random erasing augmentation (see above). No dropout, label smoothing, or auxiliary losses.
- **Did you use any pre-processing steps?** (X) Yes, ( ) No
  If yes, please detail:
  Aspect-ratio-preserving letterbox resize to 224×224 with grey (128) padding, followed by CLIP channel normalization (mean $(0.481, 0.458, 0.408)$, std $(0.269, 0.261, 0.276)$). Bilinear interpolation.
- **Did you use any post-processing steps?** (X) Yes, ( ) No
  If yes, please detail:
  Per-attribute additive bias in logit space, calibrated on the validation set by coordinate ascent maximizing the challenge metric (harmonic mean of mA and Instance_F1); 2 sweeps over a grid of 17 bias values per attribute in $[-3, 3]$. At inference: $\sigma(z_a + b_a)$, thresholded at 0.5.
- **Did you use Vision Language Models (VLMs) of any kind?** ( ) Yes, (X) No
  If yes, please detail:
  No. Although the encoder originates from a vision–language pre-training (CLIP), our pipeline uses only its visual tower as an image encoder; no text prompts, no text encoder, no zero-shot classification, and no language supervision were involved in training or inference for this track.

IV. Code repository

Link to a code repository with complete and detailed instructions so that the results obtained on Codabench can be reproduced locally. This includes a list of requirements, pre-trained models, and so on. Note, training code with instructions is also required. This is required for all participants to appear in the final ranking. **Organizers strongly encourage the use of docker to facilitate reproducibility**.

Code repository: `http://your_link_here`

References

[1] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 48th International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.